40-year annual time series of dominant tree species and leaf type cover for Italy derived from Landsat using deep learning

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Abstract. Consistent long-term maps of forest functional type and dominant tree species cover at high spatial resolution are currently unavailable for Italy. The study presents annual 30 m resolution maps of continous evergeen broad-leaved, deciduous, and coniferous leaf type cover fractions and dominant tree species class probabilities for all forested pixels in Italy spanning 1987–2025, derived from the Landsat Collection-2 Level-2 archive (Landsat 4/5/7/8/9). Two Time Series Transformer models are trained via a three-stage workflow combining contrastive pretraining with supervised fine-tuning on Italian forest plot observations, forest inventories, and a synthetically generated training dataset to predict both tasks. Pixel-level uncertainty estimates are provided as the standard deviation of a five-seed deep ensemble for both models. A three-year observation pooling strategy is applied to increase cloud-free observation density and mitigate inter-annual variability. Evaluation of the time series maps are based on hold-out areas and a comparison with historical orthophotos. The 39-year time series enables historical reconstruction of forest composition dynamics and long-term ecological change detection at national scale and relevant time scales for climate-driven species range shifts.

1 Introduction

Landsat for long term forest monitoring: With nearly 40 years of continuous acquisitions, the Landsat archive constitutes the longest consistent Earth observation record available for global vegetation monitoring (Wulder et al., 2022). Its four decades of continuity makes it uniquely suited for change detection or hindcasting of environmental variables including forest habitat characteristics (Bell et al., 2024; Turubanova et al., 2023). The archive has been applied to discriminate forest types and quantify forest variables such as canopy height or greening trends from regional to continental scales (Pflugmacher et al., 2019; Obuchowicz et al., 2024; Yel et al., 2026). In monitoring long-term changes and disturbances in forest vegetation under climate change, the Landsat afchive provides a valuable modelling base (Viana-Soto and Senf, 2025; Grünig et al., 2026). The relatively slow change and succession dynamics of forest vegetation, which is unfolding over decades, make this long temporal availability particularly valuable. A shorter observation period would fail to capture natural stand-level transitions such as species shifts or forest densification, which happen gradually over long time windows (Tong et al., 2023; Midolo et al., 2026). The medium spectral, spatial, and temporal resolution of Landsat (6 reflectance bands, 30 m ground sampling distance, and 8–16 day revisit time) is adequate for capturing annual phenological dynamics at stand level, which demonstrated to be critical for distinguishing tree species and functional leaf types (Kollert et al., 2021; Hemmerling et al., 2021; Wang et al., 2022).

Challenges: Despite these strengths, the use of multi-decadal Landsat data for vegetation mapping raises several methodological challenges. The number of usable cloud-free observations has increased since the early periods of the archive, due to the addition of new Landsat satellites. This systematic increase in observation density has been shown to cause spurious trends in annual maximum NDVI, where up to 50% of observed Landsat-based greening in alpine environments can be attributed to observational sampling bias rather than true vegetation change (Bayle et al., 2024). A second challenge arises from the failure of the Scan Line Corrector (SLC-off) on Landsat 7 ETM+, which introduces systematic spatial data gaps. Finally, spectral radiometric differences between sensors — arising from different sensors, band widths, and data processing procedures across Landsat missions — introduce inter-sensor discontinuities that may produce spurious changes in modelled variables across the 40-year record.

ML/DL for dominant tree species and leaf type cover mapping: Multitemporal satellite data consistently outperform single-date or mosaic-based approaches for tree species and leaf type mapping. Phenologically informative seasons — particularly leaf-off periods and autumn senescence — contribute disproportionately to species discrimination (Kollert et al., 2021; Tan et al., 2025). When sufficient temporal coverage is available, dense time series exploiting the full phenological cycle can outperform the inclusion of additional environmental predictor variables such as topography or climate indices (Hemmerling et al., 2021). These findings highlight temporal resolution and coverage as primary sources for improving species-level modelling and mapping performances.

Deep learning, and in particular Transformer-based architectures, provide a framework for processing raw, irregular, multi-band satellite time series (STS). Transformers process sequences of observations without requiring interpolation to a regular sequence, an essential property for long Landsat archives with varying observation density across decades and sensor platforms (Zerveas et al., 2020; Yuan et al., 2022). Self-supervised pretraining on large unlabelled STS or image time series datasets has been shown to improve representation quality and performance on downstream tasks, particularly in data-scarce settings (Yuan Yuan et al., 2020; Hiebl et al., 2025; Feng et al., 2026). Contrastive learning approaches exploit temporal and seasonal variability by training representations to be invariant across different observation windows of the same location — a property suited to stable land cover types such as forest stands (Mañas et al., 2021; Chen et al., 2020). Providing calibrated uncertainty estimates alongside predictions has emerged as an important requirement for ecological mapping applications, to enhance interpretability by users. Deep ensembles of independently trained networks deliver calibrated epistemic uncertainty that reliably identifies out-of-distribution inputs and unreliable predictions, when feature and label related domain shifts appear during the mapping phase (Sylvain et al., 2024; Hiebl et al., 2025).

Relevance for Italys forests: Italian forests are undergoing rapid and spatially complex changes. The Italian National Forest Inventory (INFC2015) documents a continuing expansion of forested area from approximately 10.46 million ha in INFC2005 to 10.98 million ha in INFC2015 driven by land abandonment and targeted reforestation programmes (Gasparini et al., 2022). At the same time, land abandonment in combination with climate change alters phenological characteristics and forest composition in favor of species with higher drought resilience (Herraiz et al., 2025; Chelli et al., 2017), while evergreen broad-leaved species are spreading in deciduous forest areas (Conedera et al., 2018). These processes are spatially heterogeneous, reflecting regional variety in land use changes, climate change, and historical species composition. Despite these dynamics, spatially explicit maps of forest composition and variables in Italy remain restricted to static maps with no continuous temporal resolution. The most recent national reference product, the Carta Forestale d'Italia 2020 (CFI2020), provides a single-epoch wall-to-wall forest map at 1:10,000 scale based on photo-interpretation of aerial orthophotos, but does not capture interannual dynamics or historical trajectories of change (Mattioli et al., 2025).

What will be done? Our study will fill this gap by providing data-driven, spatio-temporal continuous maps of leaf type cover (LTC) fractions and dominant tree species (DTS) probabilities for all forested areas in Italy - annually and at Landsat-native $30m$ resolution. We assemble and process 40-years of Landsat Collection-2 Level-2 time series from 1985 to 2025 (Landsat-4/5/7/8/9) and train multiple Time Series Transformer models (TSTpad) via contrastive pretraining and a fine-tuning workflow. The final product are annual $30m$ maps of i) 11 dominant tree species classes, and ii) LTC fractions for evergreen broad-leaved (EVE), deciduous broad-leaved (DEC) and coniferous (CON) functional leaf type. Additional ensemble-based uncertainty estimations will be provided per-pixel and target variable.

2 Materials and Methods

2.1 Study Area

The study covers all forested areas of peninsular and insular Italy, spanning several biogeographical, climatic, and latitudinal regions and an altitudinal range from sea level to approximately $2500\,m.s.l.$ (**?**) in the Alps (Fig.1). Italian forests fall within alpine, temperate, sub-Mediterranean, and Mediterranean climate zones (Chelli et al., 2017), and are dominated by four forest habitat types: broadleaved-

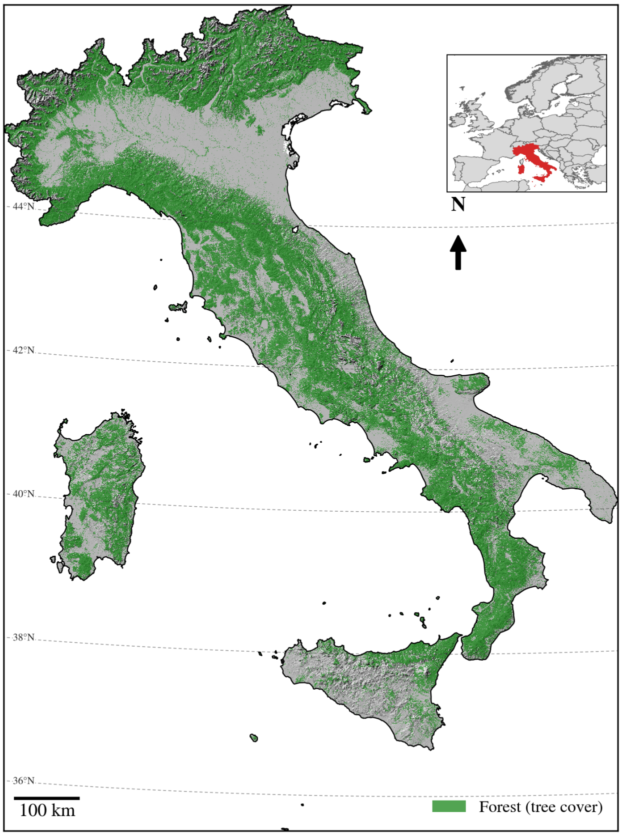

Figure 1. Study Area Italy with high forest cover in the Alps and mountainous areas of Central Italy.

deciduous and temperate coniferous forests concentrated in the Alps and Apennines, and evergreen broad-leaved and Mediterranean coniferous forests concentrated along the coasts and in the south (Bricca et al., 2026). This climatic and ecological heterogeneity is a central challenge for wall-to-wall mapping, as models trained predominantly on well-sampled regions risk out-of-domain predictions in climatically and ecologically different areas with sparser reference data. Forests cover 37% of Italy's territory (10.98 million ha) (Gasparini et al., 2022). Italy's position as a transition zone between the Mediterranean and the temperate climate of Central Europe makes its forests especially vulnerable to climate change: warming and increasing drought frequency are already documented to shift phenology and favour more drought-resilient and evergreen broad-leaved species at the cost of deciduous ones (Chelli et al., 2017; Herraiz et al., 2025; Conedera et al., 2018). These climate-driven shifts compound ongoing changes in forest management and land use, jointly reshaping tree species distribution and biodiversity across the country (Bricca et al., 2026).

2.2 Modelling Data

An overview of the used datasets during training are provided in Tab.1 and Fig.2.

2.2.1 Italian forest vegetation plot databases

The Forest Vegetation Database Italy (VDBI) is a compilation of over 16,000 phytosociological and vegetation relevé plot observations collected across Italy from the 1970s to the present. Each plot record contains species composition data, structural attributes, and cover estimates for the tree strata (Alessi et al., 2019). The VDBI covers the full range of Italian forest types from Mediterranean macchia to subalpine conifer belts. It provides temporal depth extending back to approximately 1972, enabling historical training data for pre-2000 Landsat time steps. The location accuracy of single plot observations in the dataset varies across observation year and source dataset. To avoid non-forest pixels leak into the training and test datasets, VDBI was cleaned manually based on orthophoto imagery. The final dataset after the cleaning process provides 9837 plot observations. Plot records include LTC estimates (EVE, DEC, CON) and tree species cover fractions, which were used to retrieve dominant DTS classification labels, allowing joint training of the regression ($VDBI_{DTS}$) and classification ($VDBI_{DTS}$) TST-pad model. A 10×10 km regular polygon grid was created for the train/validation/test split to reduce spatial autocorrelation between training and validation sets. A 20% validation split, stratified by polygon identifier is applied throughout all training stages (Safonova et al., 2023).

2.2.2 Field plot observations

The Vegetation Plot Observations (VPO) dataset consists of recent field campaigns conducted within 7 Italian National Parks or protected areas (see: Hiebl et al. (2025, 2026)). Each site comprises four systematic 100 m^2 vegetation plots arranged on a regular grid, within which species-wise cover is estimated for three vertical layers (0–1 m, 1–10 m, and above 10 m). LTC is extracted as sum of EVE, DEC, and CON species cover fractions, while DTS is - similar to VDBI - retrieved from species cover by selecting the tree species with highest cover fraction. The VPO data used in this study consists of observations from 2023 to 2025 field season, yielding a total of 2016 individual plot observations from 504 sites. The VPO dataset provides high-quality, spatially clustered reference data for the most recent Landsat time steps (post-2020) and serves as the primary validation reference for model performance in the recent high-density observation period. The VPO dataset is further split into a training set (80%) and a validation set (20%) to evaluate model performance. The split was performed spatially after clustering the sites by their coordinates into 10 clusters per region. The spatial clustering was done to stabilize training and prevent

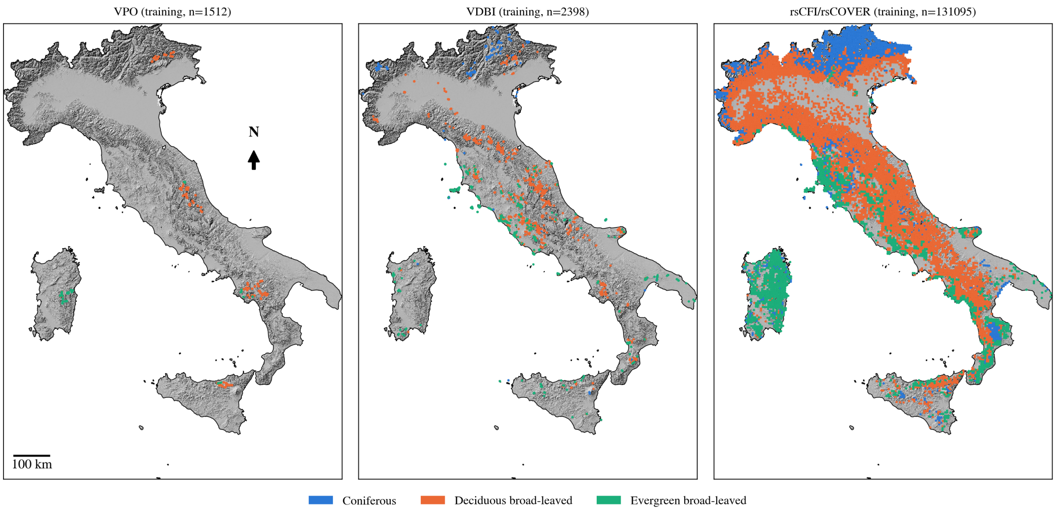

Figure 2. Overview of training datasets used in the study.

Table 1. Overview of datasets used for model training and validation.

	n samples	LTC	DTS	
VPO	2016	yes	yes	conducted in the field 2023-2025, standardized 10×10 m plots (see Sec. 2.2.2)
VDBI	2998	yes	yes	observations with varying size, acquisition time, and location accuracy ($0.01 - 10$ km) (see Sec. 2.2.1)
rsCFI	131095	-	yes	randomly sampled across CFI dominant tree species (see Sec. 2.2.3)
csCFI	30380	-	yes	pure stand polygons sampled from CFI (see Sec. 2.2.3)
rsCOVER	131095	yes	-	same locations as rsCFI, synthetically created using AlphaEarth embeddings (see Sec. 2.2.4)
csCOVER	30380	yes	-	same locations as csCFI, synthetically created using AlphaEarth embeddings (see Sec. 2.2.4)

over-optimistic accuracy estimates due to spatial autocorrelation (Safonova et al., 2023; Hiebl et al., 2025).

2.2.3 CFI Forest Inventory Data

The CFI2020 is the first national forest map of Italy produced at operational scale (1:10,000), based on photointerpretation of aerial orthophotos from 2018 to 2020 against three simultaneously applied forest definitions (Mattioli et al., 2025). The CFI2020 delineates forested areas across all 21 Italian regions and provides stand-level attributes including forest category, silvicultural system, and DTS group. The initial 43 unique classes in the dataset were collapsed into 11 dominant tree species classes: silver fir, spruce, pine, deciduous oak, chestnut, beech, manna ash - hornbeam, evergreen oak, coniferous others, deciduous others, evergreen others. In this study, the CFI2020 polygon layer serves two roles: i) as an additional training/validation data source for the DTS classification task, and ii) as a source of decision rules for cleaning the synthetic LTC training dataset (see 2.2.4). In the first case two different datasets are created by i) randomly sampling a set of evenly distributed points over all classes (rsCFI), and ii) by sampling points from manually delineated pure stand polygons that are evenly distributed across classes (csCFI). In the second case, CFI2020 species attribution (DTS) is used to impose ecologically plausible constraints on predicted cover fractions of the synthetic dataset (see 2.2.4).

2.2.4 Synthetic leaf type cover data

Spatially explicit wall-to-wall leaf type cover data for model training are not available for Italy. Following the approach of Kang et al. (2021) and Tan et al. (2025) — who demonstrate the use of coarser or indirectly derived data sources as proxy training targets where direct measurements are absent — an artificial leaf type cover dataset was created from the VDBI and VPO plot observations. A Random Forest regression model was trained on the combined VDBI and VPO training split, with EVE, DEC, and CON cover fractions as prediction targets. As input features, AlphaEarth embeddings (Brown et al., 2025; Hiebl et al., 2026) — multimodal spectral-spatial-temporal feature representations derived from a globally pretrained geospatial foundation model — were extracted for all training plot locations. AlphaEarth embeddings were shown to achieve high species classification accuracy with very limited training data in Italian mountain forests (Ball et al., 2026), making them particularly suited as inputs for the pseudo-label generation step.

The trained Random Forest model was then applied to the full CFI2020 forest extent to generate a nation-wide pixel-level pseudo-label layer for each cover fraction. The labels were extracted at the same locations as rsCFI (rsCOVER)and csCFI (csCOVER). This pseudo-label layer constitutes the primary continuous-cover target during the contrastive pre-training and early fine-tuning stages. To reduce label noise introduced by model prediction errors, the CFI2020 species attribution was used to apply ecologically motivated cleaning rules: e.g. pixels within pine-dominated forest polygons were required to have at least 60% CON cover and fewer than 30% EVE cover; analogous rules were applied to classes of beech, oak, and dominant tree species. These cleaning rules remove the most pseudo-label errors while preserving the spatial completeness of the training dataset. If a sampled synthetic plot does not meet the criteria the sample is dropped. This way we prevent ecologically non-feasible labels being forwarded into the final training loop.

2.2.5 Landsat time series

figure: landsat time series plot over all years for 2 different dominant tree species, include modelling approach with 3 year window aggregation

Annual Landsat time series were assembled for the full Italian national territory from Microsoft Planetary Computers STAC Catalog using Landsat Collection-2 Level-2 surface reflectance products from Landsat 4 TM (1982–1993), Landsat 5 TM (1984–2013), Landsat 7 ETM+ (1999–2023), Landsat 8 OLI (2013–present), and Landsat 9 OLI-2 (2021–present). Scenes with cloud cover higher than 75% were excluded, which led to a per year count of approx. 80 to 120 scenes. Per-pixel cloud, cloud shadow, cirrus, and snow masking was performed using the Landsat Collection-2 Level-2 QA PIXEL quality bitfield. Per-observation quality flags derived from QA PIXEL are propagated through the entire modelling pipeline as a key-padding mask, ensuring that cloud-contaminated or padded time steps are ignored by all attention layers of the TSTpad model. For each forested pixel, 14 spectral features are computed at each valid observation: six surface reflectance bands (blue, green, red, near-infrared, SWIR1, SWIR2) and eight derived spectral indices (NDVI, NIRv, NDMI, EVI, WDRVI, NBR, NDWI, GNDVI). At the training plot location the resulting multi-annual time series are stored as Zarr-compressed arrays to enable efficient random-access loading during model training. For mapping, the Landsat scenes were fetched and preprocessed on-the-fly in 25×25 km chunks from Microsoft Planetary Computer parallel to the GPU-based model prediction.

2.2.6 Additional data sources

The ESA WorldCover product is used as an initial broad-scale delineation of forested areas across Italy (Zanaga et al., 2022). Only pixels labelled as tree cover are forwarded for prediction in the final maps. We are aware that the ESA WorldCover product does not provide a sufficient forest mask for detailed analysis of the datasets, as many mixed pixels and edge cases are remaining. But by applying a conservative masking during mapping we leave room for stricter forest delineation through other products without loosing regional accuracy.

2.3 Time Series Transformer Architecture

The backbone model is an encoder-only Time Series Transformer extended to support irregular, sparse, annual satellite image time series (TSTpad). The architecture employs multi-head self-attention across the temporal dimension, enabling the model to learn both short-range phenological features and long-range inter-seasonal dependencies within the aggregated 3-year observation window (Rußwurm and Körner, 2020; Zerveas et al., 2020).

For each target year, three consecutive years of Landsat observations are concatenated, encompassing the target year and the two immediately preceding years (year-2, year-1, year), and aggregated into a single annual cycle to densify the sparse annual time series. After aggregation, observations are chronologically sorted and padded to a fixed sequence length of 200 time steps. Positions corresponding to padded time steps receive a key-padding mask that prevents these positions from contributing to attention score computations (Vaswani et al., 2023). The actual acquisition day-of-year for each observation is encoded as a sinusoidal positional encoding, enabling the model to correctly represent the irregular temporal spacing of Landsat acquisitions and to distinguish phenological phases across the observation window without requiring interpolation to a regular grid (Yuan Yuan et al., 2020). Input features are normalised per band using a robust standardisation based on the median and interquartile range (quantiles q=0.1 and q=0.9) — with transform parameters fitted on the training data and frozen at inference time to prevent data leakage. A sensor platform embedding with $sensordim = 5$ encodes a multi-hot vector per year indicating which Landsat satellites (L4, L5, L7, L8, L9) contributed to the observations. This embedding is added to the pre-attention temporal encoding via a learned linear projection and makes the Transformer aware of inter-sensor radiometric and observation density differences over the 40-year archive. A post-attention pooling layer using a learned query vector collapses the temporal dimension into one single summary vector per output target, letting each targets learned query decide, which valid timesteps matter most for its prediction. In a setting with target classes from a wide range of phenological and spectral characteristics, this pooling strategy allows the model to learn class-specific temporal patterns across the observation window. The model heads are two-layer Multi-Layer-Perceptrons including Dropout and GeLU activation functions (Hendrycks and Gimpel, 2016). The classification model heads map the Transformer's output to class probabilities over eleven dominant tree species and three functional leaf type classes trained with two separate Cross Entropy Loss (CELoss) computations summed into a single loss value (dominant species and functional leaf type). With the dual loss the model learns to simultaneously capture species identity without losing the ability to capture the distinction between EVE and CON leaf types. The regression heads trained with Mean Squared Error Loss (MSELoss) maps the output to continuous EVE, DEC, and CON cover fractions.

2.4 Training procedure

TODO workflow figure.

The model is trained via a three-stage progressive workflow designed to maximise temporal stability and to accommodate the heterogeneous quality and temporal coverage of available reference data.

Stage 1 - Contrastive pretraining. The backbone is trained using a joint loss that simultaneously optimises a supervised classification objective (DTS or LTC) using CELoss/MSELoss and an unsupervised normalised temperature-scaled cross-entropy contrastive loss (NT-Xent) (Chen et al., 2020): $L = \lambda \cdot \text{NT-Xent} + \text{CE}$. For the contrastive term, positive pairs are formed from the same forest plot observed in two independently sampled three-year windows ($year_0$ and $year_1$ drawn at random), representing different but ecologically equivalent observation contexts for the same location. NT-Xent pushes the backbone representations of these positive pairs toward each other in embedding space while repelling representations of observations from different plots. This objective forces the backbone to learn year-invariant spectral-temporal features that are not confounded by observation density or inter-annual radiometric variability (Mañas et al., 2021). Positive pairs are further diversified by time series augmentation, which applies three independent perturbations to each view: random temporal masking (up to 20% of valid time steps zeroed), spectral noise injection (additive Gaussian noise with $\sigma = 0.01$), and band masking (10% of spectral bands zeroed per sample) (Yuan et al., 2025). For contrastive pretraining only data from csCFI and csCOVER are used, as these pure forest stands are considered stable over time. The contrastive and classification signals are routed through separate heads attached to the shared backbone: a two-layer MLP projection head for NT-Xent, and the classification/regression head for CELoss or MSELoss, respectively. The model is optimised using AdamW with a cosine learning rate schedule preceded by a 10% warm-up phase. Stage 1 training runs for 60 epochs with a batch size of 128. Reference data consist of csCFI, rsCFI, csCOVER, and rsCOVER. Class imbalance is addressed jointly by class-weighted CrossEntropyLoss and by sample weights set to the square root of the inverse class frequency, following the approach of Grabska-Szwagrzyk et al. (2024). A stratified 20% validation hold-out, stratified by plot polygon identifier, is maintained across all training stages to prevent spatial autocorrelation between training and validation sets.

Stage 2. Fine-tuning on labelled data. The contrastive backbone from Stage 1 is loaded and fine-tuned under a purely supervised objective (CrossEntropyLoss + MSELoss). The embedding layer is kept frozen in this stage to preserve the contrastive representations, while the task heads are retrained using a reduced learning rate to prevent catastrophic forgetting (Hiebl et al., 2025). This stage runs for 10 epochs on the full 1985–2025 Landsat time series, allowing the model to adapt the task heads to the full historical archive without disturbing the temporally invariant representations established during contrastive pretraining. The used reference data are the VDBI historical plot observations.

Stage 3. Fine-tuning on plot observation data. Additional fine-tuning passes are applied from the Stage 2 checkpoint giving priority to recent high-quality VPO field observations that provide the most accurate ground reference data for the present-day forest state. This stage is particularly intended to anchor the model's predictions of recent (post-2020) DTS and LTC to field-validated reference values, while the contrastive representations established in earlier stages ensure that this anchoring remains consistent with historical predictions.

Five independent model instances, differing in random seed, architecture micro-choices within TSTpad, and random subsets of training and validation data, are trained per stage. The five-seed ensemble delivers two outputs per pixel per year: the ensemble mean prediction (used as the primary data product) and the inter-seed standard deviation (used as the pixel-level uncertainty estimate). High inter-seed uncertainty identifies pixels where the model predictions are unstable, typically in regions of low observation density, unusual spectral composition, or high class confusion.

2.5 Mapping scheme

The forest prediction domain is defined by the ESA WordCover forested area (see 2.2.6). For each target year from 1985 to 2025, the corresponding three-year Landsat observation window (year-2 to year, inclusive) is assembled from the Landsat Collection-2 Level-2 archive, cloud-masked, and processed to the 14 spectral features described in Section 2.2.5. Features are normalised with the fixed training-time standardization parameters and padded to 100 time steps. The five trained ensemble models are applied independently to each forested pixel, and the resulting class probability distributions and cover fraction predictions are aggregated across ensemble members to produce the mean prediction and inter-member uncertainty layers.

Observation density varies substantially across the archive. Before 1999, when only Landsat 5 TM was operational, the three-year window typically yields fewer cloud-free observations than the post-2013 period when Landsat 8 and 9 operate simultaneously. This density gradient is an analogue of the observational sampling bias documented for NDVI trend analyses by Bayle et al. (2024): predictions in the early archive period carry higher uncertainty due to sparse temporal sampling, which may cause the model to have lower confidence in phenological-season attribution.

We did not process the resulting annual time series using outlier detection or temporal smoothing, before storing the published LTC and DTS prediction maps. This was a design choice so downstream users can - and are encouraged to - apply temporal smoothing or outlier detection methods specifically appropriate to their application context. This is a necessary step for downstream tasks, as the time series contain both true ecological change, but also spurious noise from observation gaps, cloud contamination, and inter-annual radiometric variability.

2.6 Evaluation

The validation strategy is designed to assess both the accuracy of the map product at recent time steps and the temporal consistency of predictions across the full 40-year archive. Three independent validation datasets are employed. The VDBI hold-out test split provides the primary historical validation reference, with plot observations dating back to approximately 1995. This split allows accuracy assessment at historical time steps (1995–2025) where field data are available, spanning the Landsat 5-only and Landsat 5/7 overlap periods. The VPO hold-out split — collected during the 2023 to 2025 field work — provides the primary validation reference for recent predictions (post-2020) from the high-density Landsat 8/9 period. This dataset contains data from many mixed forest stands with hard-to-predict labels. The two validation datasets provide complementary temporal coverage: the VDBI split quantifies historical accuracy under lower observation density, while the VPO split quantifies accuracy under the best available observation conditions. A third validation source consists of a set of pure-stand forest polygons manually interpreted from high-resolution Google Earth Pro imagery in combination with the CFI dataset at recent time steps. These polygons, selected to represent spectrally unambiguous mono-specific stands, enable validation of the classification output under conditions where the expected dominant species label is highly reliable, providing an upper bound on classification accuracy unconstrained by label uncertainty in mixed stands or uncertain plot observation coordinates.

For the classification head, overall accuracy (OA) and per-class F1 score are computed for each period. For the regression heads, root mean squared error (RMSE) and mean absolute error (MAE) are reported for each of the three cover fractions (EVE, DEC, CON). The VPO dataset is finally used to assess in detail the model's ability to predict DTS and LTC in mixed stands, where the prediction of a single label is inherently more difficult and uncertain.

To account for uncertainties in plot observations coming from (i) a spatial mismatch between plot location and Landsat pixel, and (ii) temporal uncertainty and variations between plot observation year and Landsat acquisition year, we complement the direct accuracy metric with a true-label-conditioned metric (Lobo et al., 2020). Predictions are evaluated over a space-time neighborhood around each plots reported coordinates and year, weighting candidate locations by a Gaussian space-time prior (location and temporal uncertainty) combined with a Gaussian likelihood that down-weights candidates whose predicted value strongly disagrees with the plots true label, following the logic of likelihood-based geolocation correction (Shannon et al., 2024). This estimator is bracketed by two limiting cases obtained by varying the likelihood bandwidth: a pure prior-weighted neighborhood mean that ignores the label entirely, and a best-match estimate that selects the single most consistent candidate location. We report accuracy metrics (RMSE, MAE and R^2) computed under all three methods to characterize how sensitive model accuracy is to possible location and temporal errors (see Appendix A).

3 Results and Discussion

The final maps provide following variables per pixel and per year (1987 to 2025):

- Dominant tree species: ensemble mean of dominant tree species class and level of agreement across ensemble members; ensemble mean of class probabilities and ensemble standard deviation for 11 classes
- Leaf type cover fractions: ensemble mean of EVE, DEC, and CON cover fractions and ensemble standard deviation

3.1 Performance metrics on test datasets

TODO.

- conventional metrics for all datasets

The test split of the recently conducted, homogenuos VPO dataset and the test split of the multi-temporal VDBI collection dataset are used to assess the error of the produced multi-temporal LTC maps (Tab. 3). The VPO plot observations have high location accuracy compared to the 30 m Landsat resolution and we assume that the direct plot location is a fair metric here. The maps show an MAE of $12.5\%_{cover}$, $9.7\%_{cover}$ with an R_2 of 0.66 and 0.86 for EVE and CON on the VPO test dataset. The absence of plots with coniferous tree species in the VPO dataset makes interpretation of performance metrics of CON cover obsolete.

The error increases significantly with the VDBI dataset and historical plot observations. Interpreting results in the context of historical plot observations with low accuracy or no GNSS measurement is difficult. Nonetheless we think reporting the error despite the high spatial bias is valuable.

In pure stands with one of EVE, DEC or CON cover converging towards $100\,\%_{cover}$, the error drops significantly compared to mixed forest stands with e.g. equal amounts of EVE and DEC cover. Other studies report similar difficulties in predicting mixed forest stands for differing target variables source missing. However, despite the high amount of mixed forest stands contained in the VPO dataset reasonable MAE and RMSE, in line with other studies, were achieved by the model. Regarding the error estimation on the VDBI dataset a more cautious approach is necessary. Directly measuring the error at plot locations that might be several hundred meters off its true location inflates error metrics significantly. But with higher amounts of plots in coniferous stands with comparatively low error rates - also in regard to the best-match and importance-weighted error estimation - and good performance on the recently conducted VPO dataset

- metrics for leaf type classes in classification task

- probabilistic metrics for all datasets
- comparing pure and mixed stands

- confusion matrix for interpretation very important (confusing classes that are often mixed forest stands)

3.2 Dominant tree species dynamics

TODO. - figure of transitions of 3 most likely transformations - figure of biggest areas of change - table per class of highest losses/gains

3.3 Leaf type cover dynamics

TODO. - figure of theil sen slopes of the 3 target classes for all of italy - table of forest type/habitat type with highest changes

3.4 Limitations and future work

TODO.

- label noise and location accuracy issues (Persson et al., 2022; Strunk et al., 2025) in recent and historical plot observations. This affects the training massively, but also the test time performance reporting. Applying different computation techniques for metrics tries to tackle that but it does not compensate for reliable location measurements of independent test plots.
- inter-annual spectral noise that leads to inconsistencies in per-pixel time series, assumption is a stable state or a single disturbance/gradual succession change of states. Random fluctuations can be considered noise. A Hidden Markov model + Vertibri smoother could help in smoothing the per-year argmax predictions and stabilize the probability transition vectors of each classes.
- need for a harmonized full scale Landsat archive as analysis ready product (sensor calibration, sensor merge, density differences etc.)
- processing is huge limitation, time series into deep learning models (several weeks of mapping)
- foundation models for landsat time series (torchgeo) or/and embeddings -> easy/fast to use for processing long time series on large areas

4 Conclusions

The Landsat collections provide an unprecedented archive of satellite imagery for reconstructing and hindcasting environemntal variables across large regions. Few studies exist, that exploit the full 40 years of available Collection-2 imagery to map forest related variables and provide time series datasets, that can be used to investigate forest dynamics and changes under climate and landuse change conditions. This study predicts annual maps of dominant leaf type and leaf type cover from Landsat time series, forest plot observations and forest inventories.

5 Code and data availability

All python code that was used during data processing, training and mapping is available under https://git.uibk.ac.at/c7161037/ls_mapping (training) and https://git.uibk.ac.at/rslab/sattstools (beta version, remote sensing data processing). The model architecture and model weights can be found on "link text". The Zarr-compressed annual maps of leaf type cover fractions and dominant tree species probabilities are available on "link text".

Appendix A: Probabilistic Evaluation metrics

TODO.

- figure: sensitivity study for sigmaobs in the posterior method. lower bound direct, upper bound best match, line in between with sigma
- figure of posterior landsacape for 2 example plots. prior (gaussian) * likelihood => posterior

Appendix B: Comparison to historical orthophoto imagery

TODO.

Author contributions.

Table 2. Classification performance (precision, recall, f1-score) for the dominant tree species classification task, evaluated on three test datasets. Reported is the likelihood-based importance-weighted performance result based on the location accuracy of the plots (see Sec. 2.6).

	VPO test split				VDBI test split				CFI test dataset			
class	prec.	recall	f1-score	support	prec.	recall	f1-score	support	pre.	recall	f1-score	support
silver fir	-	-	-	-	-	-	-	-	0.86	0.85	0.86	518
other dec.	0.52	0.55	0.53	112	0.23	0.81	0.36	27	0.37	0.85	0.52	240
dec. oak	0.88	0.46	0.60	127	0.70	0.40	0.51	123	0.67	0.61	0.64	1137
chestnut	0.96	0.88	0.92	26	0.89	0.53	0.67	30	0.67	0.74	0.70	817
other con.	-	-	-	-	-	-	-	-	0.49	0.50	0.49	220
beech	0.57	0.91	0.68	108	0.89	0.78	0.83	120	0.83	0.65	0.73	594
eve. oak	0.65	0.64	0.65	53	0.69	0.70	0.70	110	0.86	0.79	0.82	1141
other eve.	-	-	-	-	-	-	-	-	0.89	0.81	0.85	503
pine	-	-	-	-	0.83	0.78	0.81	51	0.74	0.72	0.73	410
ornus-ostr.	0.69	0.43	0.53	51	0.38	0.49	0.43	60	0.61	0.65	0.63	695
spruce	-	-	-	-	0.75	0.4	0.52	15	0.95	0.72	0.82	367
accuracy				493				536				30985
macro avg	0.71	0.65	0.65	493	0.67	0.61	0.60	536	0.72	0.72	0.71	30985
weighted avg	0.68	0.62	0.62	493	0.71	0.62	0.64	536	0.75	0.72	0.72	30985

Table 3. Regression performance (MAE, RMSE, R^2) for leaf type cover (LTC) prediction, evaluated on two test datasets. X/X/X refers to the computation of the evaluation metrics, being directly from plot location, importance-weighting, and best-match (see Sec. 2.6 and Appendix A). MAE and RMSE are reported as $\%_{cover}$.

	VPO test split			VDBI test split		
target	MAE	RMSE	R^2	MAE	RMSE	R^2
EVE	12.5 / 11.6 / 8.5	19.1/17.5/14.9	0.66/0.72/0.79	TODO	TODO	TODO
DEC	9.7 / 8.7 / 8.4	12.4/10.6/7.9	0.86/0.85/0.92	TODO	TODO	TODO
CON	5.0 / 4.4 / 3.0	9.1/7.4/6.1	-0.05/0.31/0.54	TODO	TODO	TODO

TODO.

Competing interests.

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Disclaimer.

TODO.

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